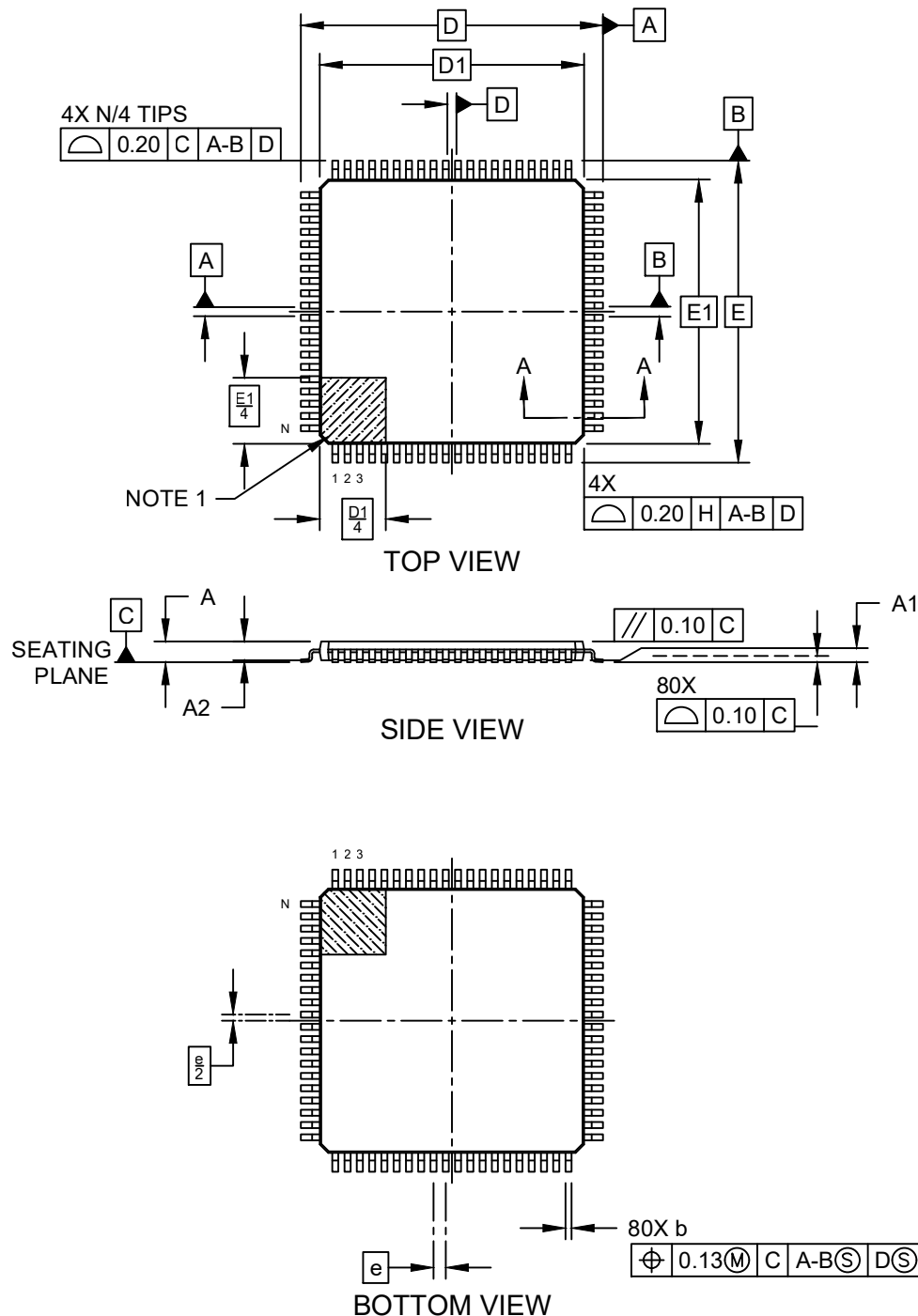


# Packaging Diagrams and Parameters

## 80-Lead Thin Quad Flat Pack (AYC) - 14x14x1.0 mm Body [TQFP]

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>

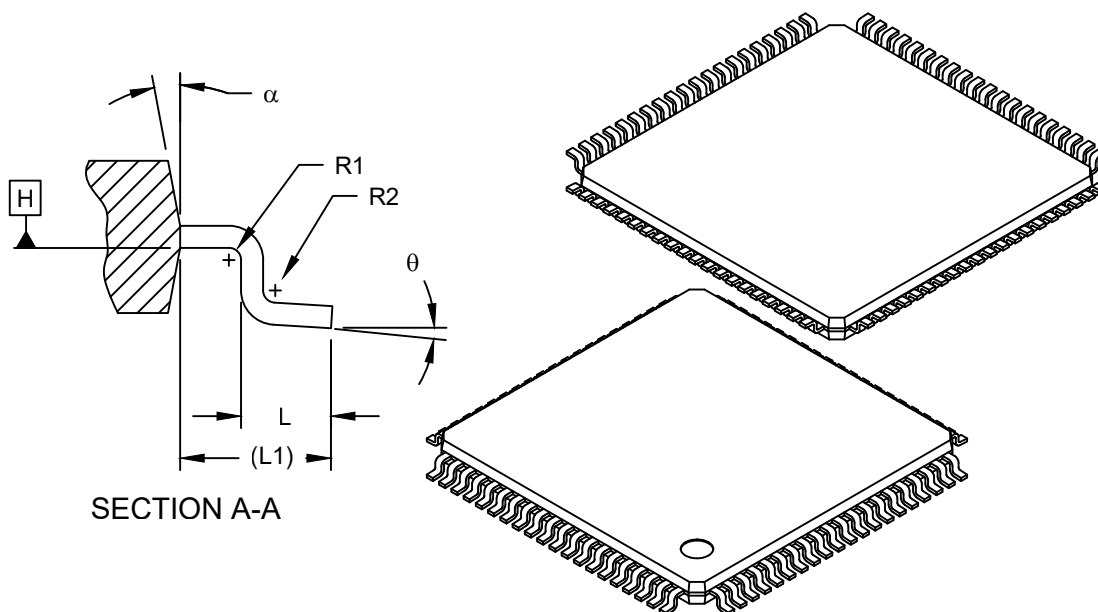


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## Packaging Diagrams and Parameters

### 80-Lead Thin Quad Flat Pack (AYC) - 14x14x1.0 mm Body [TQFP]

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|                          |    | Units | MILLIMETERS |      |      |
|--------------------------|----|-------|-------------|------|------|
| Dimension Limits         |    |       | MIN         | NOM  | MAX  |
| Number of Leads          | N  |       | 80          |      |      |
| Lead Pitch               | e  |       | 0.65 BSC    |      |      |
| Overall Height           | A  |       | -           | -    | 1.20 |
| Standoff                 | A1 |       | 0.05        | 0.10 | 0.15 |
| Molded Package Thickness | A2 |       | 0.95        | 1.00 | 1.05 |
| Foot Length              | L  |       | 0.45        | 0.60 | 0.75 |
| Lead Length              | L1 |       | 1.00 REF    |      |      |
| Foot Angle               | θ  |       | 0°          | 3.5° | 7°   |
| Overall Width            | D  |       | 16.00 BSC   |      |      |
| Overall Length           | E  |       | 16.00 BSC   |      |      |
| Molded Package Width     | D1 |       | 14.00 BSC   |      |      |
| Molded Package Length    | E1 |       | 14.00 BSC   |      |      |
| Lead Width               | b  |       | 0.22        | 0.32 | 0.38 |
| Mold Draft Angle Top     | α  |       | 11°         | 12°  | 13°  |
| Lead Shoulder Radius     | R1 |       | 0.08        | -    | -    |
| Lead Foot Radius         | R2 |       | 0.08        | -    | 0.20 |

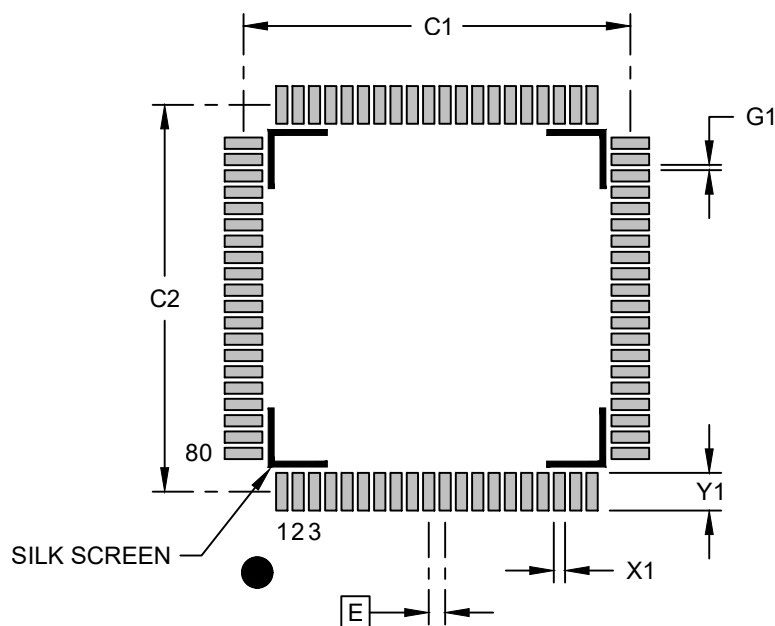
**Notes:**

- The Pin 1 visual index feature may vary, but it must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M
  - BSC: Basic Dimension. The theoretically exact value is shown without tolerances.
  - REF: Reference Dimension, usually without tolerance, is for information purposes only.

## Land Pattern

### 80-Lead Thin Quad Flat Pack (AYC) - 14x14x1.0 mm Body [TQFP]

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



### RECOMMENDED LAND PATTERN

| Units                            |    | MILLIMETERS |       |      |
|----------------------------------|----|-------------|-------|------|
| Dimension Limits                 |    | MIN         | NOM   | MAX  |
| Contact Pitch                    | E  | 0.65 BSC    |       |      |
| Contact Pad Spacing              | C1 |             | 13.40 |      |
| Contact Pad Spacing              | C2 |             | 13.40 |      |
| Contact Pad Width (X80)          | X1 |             |       | 0.45 |
| Contact Pad Length (X80)         | Y1 |             |       | 1.50 |
| Contact Pad to Contact Pad (X76) | G1 | 0.20        |       |      |

#### Notes:

- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. The theoretically exact value is shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during the reflow process.

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